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NNDL Assignment: 03

1. In an attention mechanism: Attention weights = $[0.3, 0.4, 0.3]$, Value Vectors = $[5, 2, 6]$.
- a) compute the attention output using the weighted sum formula.
 - b) Update weights to $[0.1, 0.6, 0.3]$ and recompute the attention output
 - c) compute the difference between original and update outputs
 - d) interpret how dominant attention weights affect the information captured by the model

Attention Mechanism

Given that

$$\text{weights} = [0.3, 0.4, 0.3]$$

$$\text{Values} = [5, 2, 6]$$

- a) Attention output (weighted sum):
$$= (0.3 \times 5) + (0.4 \times 2) + (0.3 \times 6)$$
$$= 1.5 + 0.8 + 1.8 = 4.1$$
- b) Updated weights = $[0.1, 0.6, 0.3]$:
$$= (0.1 \times 5) + (0.6 \times 2) + (0.3 \times 6)$$
$$= 0.5 + 1.2 + 1.8 = 3.5$$
- c) Difference
$$= 4.1 - 3.5 = 0.6$$
- d) Interpretation

Higher weights give more importance to certain values. When one weight dominates (0.6), the model focuses more on that specific input, reducing influence of others.

2. Multi-head attention layer has 2 heads: Head 1 output: $[3, 1]$, Head 2 output: $[2, 4]$ projection matrix $W = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$.
- Concatenate the outputs of both heads
 - Apply the projection matrix W to the concatenated vector
 - compute the final projected output vector
 - Explain why multiple attention heads improve transformers representation learning.

Multi-head Attention

Given that

Head 1 = $[3, 1]$, Head 2 = $[2, 4]$

$W = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$

a) Concatenation:

$$= [3, 1, 2, 4]$$

b) Apply projection:

multiply $W \times$ vector

c) Final output

$$\text{Row 1: } (1 \times 3 + 0 \times 1 + 0 \times 2 + 1 \times 4) = 7$$

$$\text{Row 2: } (0 \times 3 + 1 \times 1 + 1 \times 2 + 0 \times 4) = 3$$

$$[7, 3]$$

d) They capture different relationships / patterns simultaneously by improving representation and context understanding.

3. A deep learning model initially has 5 million parameters.

- Apply 25% pruning and compute the remaining parameters
- Apply an additional 15% pruning on the remaining parameters
- Determine the final parameters count after both pruning steps
- calculate the overall Percentage reduction from the original

e. Briefly discuss structured vs. unstructured pruning.

Model pruning

Initial Parameters = 5,000,000

a) After 25% pruning

Remaining = 75% $\rightarrow 0.75 \times 5,000,000 = 3,750,000$

b) After additional 15% pruning:

Remaining = 85% $\rightarrow 0.85 \times 3,750,000 = 3,187,500$

c) Final Parameters

= 3,187,500

d) Overall reduction

Reduction = $5,000,000 - 3,187,500 = 1,812,500$ reduction =

$(1,812,500 / 5,000,000) \times 100 = 36.25\%$

e) structured: Removes entire neurons/filters $\rightarrow$ faster on hardware

unstructured: Removes individual weights $\rightarrow$ more flexible but less hardware efficient

4. in a knowledge distillation setup:

Teacher distribution: $[0.7, 0.2, 0.1]$, student distribution: $[0.4, 0.4, 0.2]$

True label: $[1, 0, 0]$, loss combination parameters $\lambda = 0.6$

a) compute the KL Divergence b/w teacher & student distributions

b) Compute the cross-entropy loss b/w student output & the true label

c) combine losses: $Loss = \lambda \cdot KL + (1-\lambda) \cdot CE$

d) Determine the final combined loss value.

- e) Explain why the combined loss is more effective than using only cross-entropy

Knowledge Distillation

Given that

$$\text{Teacher} = [0.7, 0.2, 0.1]$$

$$\text{Student} = [0.4, 0.4, 0.2]$$

$$\text{True label} = [1, 0, 0], \lambda = 0.6$$

- a) KL Divergence:

$$KL \approx 0.7 \ln(0.7/0.4) + 0.2 \ln(0.2/0.4) + 0.1$$

$$\ln(0.1/0.2)$$

$$\approx 0.7(0.559) + (0.2)(-0.693) + 0.1(-0.693)$$

$$\approx 0.391 - 0.139 - 0.069 \approx 0.183$$

- b) Cross-Entropy (CE)

$$= -\log(0.4) \approx 0.916$$

- c) combined loss:

$$\text{loss} = 0.6 \times 0.183 + 0.4 \times 0.916$$

$$= 0.1098 + 0.3664 = 0.4762$$

- d) Final loss:

$$\approx 0.48$$

- e) it uses both

Teacher's knowledge (soft targets) $\rightarrow$ richer information

True labels $\rightarrow$ correct supervision

$\rightarrow$ leads to better generalization than using only CE